A Few Notes on the Brachistochrone Problem

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1 Introduction

Johann Bernoulli posed the "problem of the brachistochrone" to the readers of *Acta Eruditorum* in June, 1696 [2], which asks the following question:

> Given two points A and B in a vertical plane, what is the curve traced out by a point acted on only by gravity, which starts at A and reaches B in the shortest time?

The scenario Bernoulli envisioned is depicted in Figure 1.

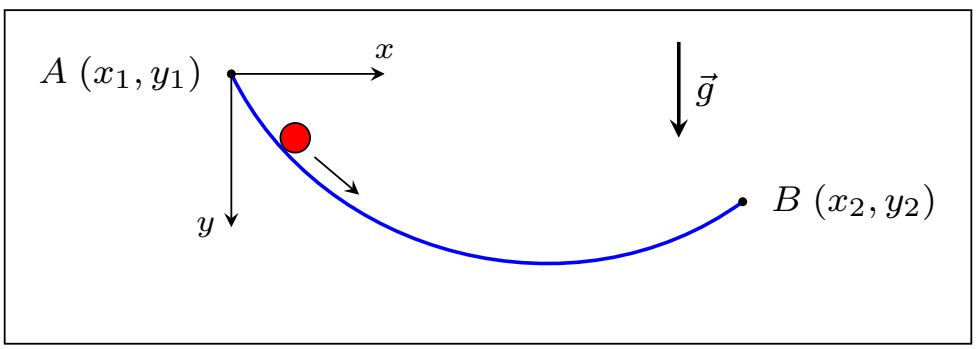

Figure 1: The Brachistochrone Problem Setup

2 A Bit of History

The problem posed by Johann Bernoulli was the called the Brachistochrone[1] problem, or the path of fastest descent. Bernoulli issued his challenge in *Acta Eruditorum* [2], writing

> I, Johann Bernoulli, address the most brilliant mathematicians in the world. Nothing is more attractive to intelligent people than an honest, challenging problem, whose possible solution will bestow fame and remain as a lasting monument. Following the example set by Pascal, Fermat, etc., I hope to gain the gratitude of the whole scientific community by placing before the finest mathematicians of our time a problem which will test their methods and the strength of their intellect. If someone communicates to me the solution of the proposed problem, I shall publicly declare him worthy of praise.

[1] The word brachistochrone comes from the Ancient Greek: *brachis* (shortest) + *chronos* (time) [13].

Bernoulli wrote the problem statement as:

> Given two points A and B in a vertical plane, what is the curve traced out by a point acted on only by gravity, which starts at A and reaches B in the shortest time.

Figure 1 attempts to capture Bernoulli's problem statement.

Galileo had attempted to solve this problem in his *Two New Sciences* [12] and had concluded, based on geometric arguments, that the solution was a circular path. Galileo's approach is shown in Figure 2. Apparently Galileo hedged a bit on his solution. In fact, it is said that he had reservations about his conclusion and suggested that a "higher mathematics" would possibly find a better solution [3].

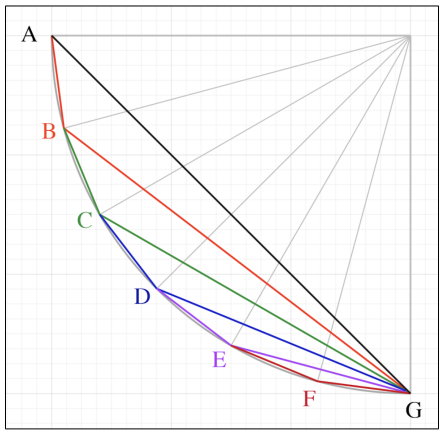

Figure 2: Galileo considered a mass falling along different chords of a circle starting at A. He proved that the path along ABG was quicker than along AG, and ABCG was quicker than ABG, and ABCDG was quicker than ABCG, etc. In this way he showed that the path along the circular arc was quicker than any set of chords. From this he (incorrectly) inferred that the circle was the path of quickest descent. Galileo held out reservations about his solution, and rightfully so.

Then in 1659, while studying the physics of pendulums and time keeping, Christiaan Huygens [8] recognized that a perfect harmonic oscillator, one whose restoring force was linear in the displacement of the oscillator, would produce the perfect time piece. Unfortunately, while the pendulum (as proposed by Galileo) was the simplest oscillator to construct, Huygens knew that it was not a perfect harmonic oscillator. In particular, he knew that the period of oscillation became smaller when the amplitude of the oscillation became larger. In order to "fix" the pendulum, he searched for a curve of equal time, called the *tautochrone* [11], that would allow all amplitudes of the pendulum to have the same period. He found the solution and recognized it to be a cycloid [9]. The cycloid is shown in Figure 3.

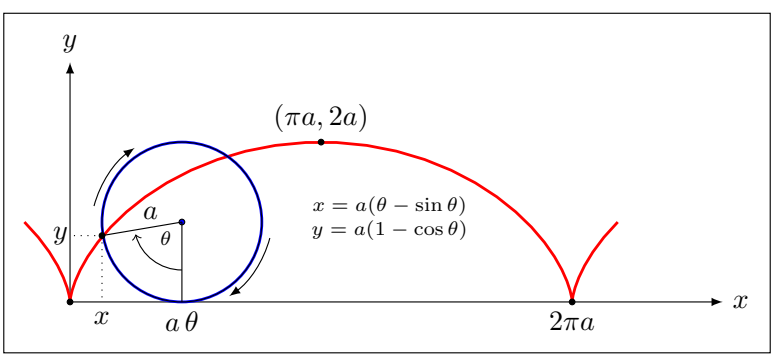

Figure 3: The Cycloid Curve

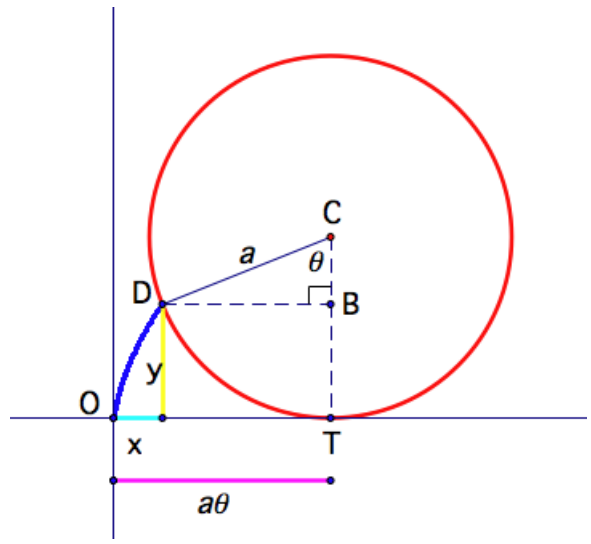

Figure 4: Finding The Coordinates Of The Point D On A Cycloid

BTW, how are the cycloid's (x, y) coordinates computed? Consider the scenario depicted in Figure 4 [4]. To find the (x, y) coordinates of the point D we look at the distances involved. We can see from the figure that

- The x-coordinate is the length of OT minus the length of DB
- The y-coordinate is the length of CT minus the length of CB (note that CT $= a$ is the radius of C)
- The length of an arc that is subtended by a central angle is equal to the radius times the central angle

Now, since the distance that circle C has rolled must be equal to the length of arc DT and the arc DT is equal to $a\theta$ we know that $a\theta =$ arc DT $=$ OT. We also know from trigonometry that DB $= a \sin\theta$ and CB $= a\cos\theta$.

If we work out these lengths we see that

$$\begin{aligned} x &= \text{OT - DB} && \text{\# See Figure 4} \\ &= a\theta - a\sin\theta && \text{\# OT = radius times the central angle} = a\theta \text{ and DB} = a\sin\theta \\ &= a(\theta - \sin\theta) && \text{\# } x = a(\theta - \sin\theta) \end{aligned}$$

and

$$\begin{aligned} y &= \text{CT - CB} && \text{\# Figure 4} \\ &= a - a\cos\theta && \text{\# CT} = a \text{ and CB} = a\cos\theta \\ &= a(1 - \cos\theta) && \text{\# } y = a(1 - \cos\theta) \end{aligned}$$

So as we saw in Figure 3, $x = a(\theta - \sin\theta)$ and $y = a(1 - \cos\theta)$.

Ok, back to the history. So almost thirty years later, during the infancy of the infinitesimal calculus, the Tautochrone problem was held up as a master example of an "optimal" solution whose derivation should yield to the much more powerful and elegant methods of the calculus. Jakob Bernoulli, Johann's brother, succeeded in deriving the Tautochrone problem in 1690 using the calculus, using the term "integral" for the first time in print, but it was not at first clear what other problems could yield in a similar way.

Then, in 1696, Johann Bernoulli posed the Brachistrochrone problem in the pages of *Acta Eruditorum*.

2.1 A (Very) Brief Timeline of Brachistrochrone Related Events

- 1638 – Galileo proposes arc of a circle as the least time solution (Figure 2)
- 1658 – Pascal puts forth a challenge involving finding the area under a segment of a cycloid
- 1659 – Huygens shows experimentally that the cycloid is the solution to the Tautochrone problem
- 1662 – Fermat's proposes the Principle of Least Time [10]
- 1696 – Johann Bernoulli poses the Brachistrochrone problem in *Acta Eruditorum* (edited by Leibniz)
- 1697 – Solutions provided by the Bernoulli brothers, L'Hôpital, Leibniz, and Newton (anonymously)
- 1707 – Euler is born
- 1726 – Newton dies
- 1726 – Euler completes his Ph.D under Johann Bernoulli
- 1755 – Lagrange, at age 19, finds an analytic solution to the Tautochrone problem
- 1755 – Euler drops his own approach in favor of Lagrange's purely analytic approach
- 1756 – Euler renames the subject the "Calculus of Variations" in his *Elementa Calculi Variationum* lecture [5]

3 The Calculus of Variations and the Brachistrochrone Problem

Recall the Brachistochrone problem setup shown in Figure 1. Here we are trying to find the *path* that takes the shortest time from point A to B. In the language of the calculus of variations [7] we want to evaluate the functional

$$I[t] = \int_0^t dt$$

Consider a curve over in which we call an infinitesimal segment of the path is ds. Then we know that the velocity of the ball (or bead, in Bernoulli's original problem statement) is

$$V = \frac{ds}{dt} \tag{1}$$

Now, treating the derivative as a fraction[2] we can see from Equation 1 that

$$dt = \frac{ds}{V}$$

and so

$$I = \int_0^t dt = \int_A^B \frac{ds}{V}$$

Next, what can se say about ds? Well, we know from Pythagoras that $ds^2 = dx^2 + dy^2$ which is shown in Figure 5.

[2] Care needs to be taken when treating a derivative as a fraction.

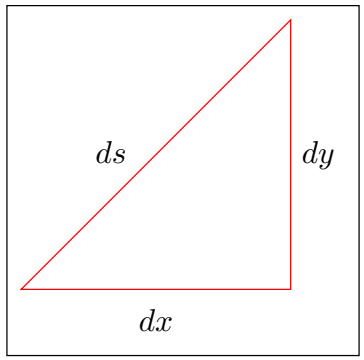

Figure 5: $ds^2 = dx^2 + dy^2$

Now we can compute ds:

$$\begin{aligned}
ds^2 &= dx^2 + dy^2 && \text{\# Pythagorean Theorem (Figure 5)} \\
&= \left(1 + \frac{dy^2}{dx^2}\right) dx^2 && \text{\# factor out } dx^2 \\
&= \left(1 + \left(\frac{dy}{dx}\right)^2\right) dx^2 && \text{\# group } \frac{dy}{dx} \\
&= \left(1 + (y')^2\right) dx^2 && \text{\# } y' = \frac{dy}{dx} \\
&\Rightarrow \quad ds = \sqrt{1 + (y')^2} dx && \text{\# note: here we're treating } \frac{dy}{dx} \text{ as a fraction}
\end{aligned}$$

Now we know that

$$ds = \sqrt{1 + (y')^2} dx \tag{2}$$

Next, we know from the Law of Conservation of Energy [1] that

$$\frac{1}{2}mV^2 - mgy = 0 \Rightarrow \frac{1}{2}mV^2 = mgy \Rightarrow \frac{1}{2}V^2 = gy \Rightarrow V = \sqrt{2gy}$$

So

$$V = \sqrt{2gy} \tag{3}$$

Now we can write

$$\begin{aligned}
I &= \int_A^B \frac{ds}{V} && \text{\# definition of } I \\
&= \int_A^B \frac{\sqrt{1 + (y')^2}}{\sqrt{2gy}} dx && \text{\# Equations 2 and 3} \\
&= \frac{1}{\sqrt{2g}} \int_A^B \sqrt{\frac{1 + (y')^2}{y}}\, dx && \text{\# pull } \frac{1}{\sqrt{2g}} \text{ out of integral, consolidate square root}
\end{aligned}$$

So we have this expression for I

$$I = \frac{1}{\sqrt{2g}} \int_A^B \sqrt{\frac{1 + (y')^2}{y}}\, dx$$

Recall that the Calculus of Variations is about finding a function that minimizes some other function. That "other function", which takes a function as an argument, is called a functional. Here I is the functional the function that is trying to be minimized, F, is

$$F = \sqrt{\frac{1 + (y')^2}{y}}\, dx$$

The Beltrami Identity [6] ...

$$F - y' \frac{\partial F}{\partial y'} = C$$

where C is some constant.

4 Conclusions

Acknowledgements

LaTeX Source

`https://www.overleaf.com/read/vpwbqqpnjywj`

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